

Design and Simulation of a PID Controller for Precise DC Motor Speed Control

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Abstract—This project details the modeling, simulation, and multi-strategy optimization of a closed-loop Proportional-Integral-Derivative (PID) controller designed for precise speed regulation of an armature-controlled permanent magnet DC motor using MATLAB/Simulink. Unregulated open-loop plant models exhibit severe speed degradation, lack state awareness, and fail to track reference trajectories when subjected to variable mechanical loads or internal thermal parameter drifts. To resolve these limitations, a high-fidelity electromechanical “Digital Twin” of the motor is derived from first-principles using Kirchhoff’s Voltage Law (KVL) and Newton-Euler angular momentum balance laws.

Three tuning paradigms are systematically analyzed and benchmarked: Heuristic Manual Tuning (P -only), Analytical Ziegler-Nichols (ZN) sustained-oscillation tuning, and Frequency-Domain Computational Auto-Tuning optimization. The transient and steady-state responses are evaluated under a step command reference of 100 rad/s and sudden external load torque disturbances. Simulation results reveal that while the Ziegler-Nichols methodology yields the fastest rise time ($T_r \approx 0.15$ s), it introduces severe current spikes and a high maximum overshoot ($M_p \approx 22\%$), presenting risks of mechanical gear stress. Conversely, computational auto-tuning achieves an optimal, critically damped response with a balanced rise time ($T_r \approx 0.65$ s), safe maximum overshoot ($M_p \approx 9\%$), fast settling performance ($T_s \approx 1.5$ s), and complete steady-state error eradication ($e_{ss} = 0\%$). The validated control loop satisfies strict robotic and CNC industrial precision requirements while ensuring structural longevity.

Index Terms—DC Motor Speed Control, Digital Twin, Kirchhoff’s Voltage Law, MATLAB/Simulink, Closed-Loop System, Ziegler-Nichols Tuning, Computational Auto-Tuning, Transient Response.

I. INTRODUCTION

Modern industrial manufacturing frameworks, including robotic joint articulators, computer numerical control (CNC) systems, and factory automation lines, demand exceptionally accurate velocity control loops capable of operating reliably under dynamic loads. Permanent magnet DC motors are widely deployed in these environments due to their linear electromechanical profiles, ease of regulation, and robust starting torque characteristics. However, in practical deployment, an unregulated actuator suffers from inherent performance degradation caused by physical limitations, including internal

brush friction, back-electromotive self-braking, and winding resistance shifts due to operating heat.

To guarantee high structural precision, a deterministic plant framework must be supplemented with a closed-loop control system. This project presents a structured electromechanical engineering layout to design, simulate, and optimize a permanent magnet DC motor system within MATLAB/Simulink. By converting the motor into a coupled multi-domain “Digital Twin,” this work establishes a rigorous mathematical foundation using Kirchhoff’s Voltage Law for the internal series circuitry and Newton’s Second Law for the mechanical rotor system.

Subsequently, a closed-loop framework driven by an error-detection summing junction is integrated into the model. Three primary controller tuning methods—Heuristic Manual Analysis, Analytical Ziegler-Nichols, and Frequency-Response Auto-Tuning—are benchmarked against transient response constraints (rise time, peak overshoot, settling time) and steady-state offset boundaries. The finalized control loop provides total trajectory tracking and immediate disturbance rejection, delivering a robust platform ready for high-fidelity mechatronic implementations.

II. PROBLEM STATEMENT

Traditional open-loop DC motor architectures follow input armature voltages blindly due to a lack of situational tracking awareness. When a mechanical load change is introduced, the motor speed drops significantly because the plant lacks an automated reference pathway to request additional potential energy from the supply source. Furthermore, internal viscous friction losses and heat generation cause parameters like winding resistance to drift continuously over prolonged periods, leading to precision deficits that make unregulated systems unsuitable for high-precision manufacturing applications.

The challenge of this project is to synthesize a mathematical model of an armature-controlled DC motor plant in MATLAB/Simulink and integrate a continuous closed-loop PID control block capable of eliminating tracking offsets. The system must overcome steady-state errors and enforce critical transient damping bounds to protect delicate internal mechanisms from severe current spikes and excessive overshoot.

Consequently, this study develops and benchmarks three distinct tuning methodologies to identify an optimal mathematical solution that ensures high accuracy, fast settling behavior, and high-fidelity disturbance rejection under dynamic operating constraints.

III. OBJECTIVES

The core objective of this project is to design, model, and optimize a closed-loop PID control architecture to manage a DC motor Digital Twin under variable operational inputs.

The explicit research objectives are:

- 1) To derive a coupled electromechanical transfer function of an armature-controlled DC motor utilizing first-principles physics.
- 2) To implement a high-fidelity plant block layout within the MATLAB/Simulink environment.
- 3) To analyze the performance of an unregulated open-loop plant under standard step commands and evaluate its vulnerability to load torque steps.
- 4) To integrate a closed-loop feedback pathway that continuously monitors actual angular velocity (ω_{act}) and generates real-time error trajectories.
- 5) To model a continuous-time Proportional-Integral-Derivative (PID) block and study the temporal focus of its individual gains (K_p , K_i , K_d).
- 6) To evaluate an experimental Manual Tuning approach (P -only) and isolate the underlying physics causing steady-state proportional droop.
- 7) To execute an analytical Ziegler-Nichols (ZN) sustained-oscillation test to establish an algorithmic control baseline.
- 8) To deploy a frequency-response computational Auto-Tuner optimized for a robust 60° Phase Margin baseline.
- 9) To benchmark all three tuning variations against rise time (T_r), peak overshoot (M_p), settling time (T_s), and steady-state error (e_{ss}).
- 10) To validate the optimal configuration's disturbance rejection capabilities when subjected to an external load torque step input.

IV. METHODOLOGY

The methodology deployed in this study covers analytical electromechanical modeling, architectural Simulink block construction, loop feedback integration, algorithmic optimization, and empirical performance verification.

A. System Architecture and Digital Twin Derivation

The physical system decomposes into an electrical armature circuit network mechanically coupled to a rotational shaft load. Speed modulation is achieved exclusively by adjusting the incoming armature input potential (V_a).

1) *Electrical Domain Formulation:* Applying Kirchhoff's Voltage Law (KVL) across the active armature circuit loop yields the core first-order ordinary differential equation:

$$V_a(t) = L \frac{di_a(t)}{dt} + R \cdot i_a(t) + e_b(t) \quad (1)$$

where $e_b(t)$ represents the electrical counter-potential (Back-EMF) generated by the rotor spinning through the stator magnetic flux lines. This phenomenon forms an electrical damping mechanism and is expressed as:

$$e_b(t) = K_b \omega(t) \quad (2)$$

2) *Mechanical Domain Formulation:* Applying Newton's Second Law to rotational assemblies establishes that the internal electromagnetic shaft torque (T_m) must balance against rotor inertia, mechanical viscous shaft damping, and external friction resistances:

$$J \frac{d\omega(t)}{dt} + B \cdot \omega(t) = T_m(t) - T_L(t) \quad (3)$$

where the raw physical torque developed is directly proportional to the active armature current moving through the motor coil lines:

$$T_m(t) = K_t i_a(t) \quad (4)$$

3) *System Transfer Function Characterization:* Applying the unilateral Laplace Transform to equations (1) through (4), assuming uniform zero initial energy states, isolates the complete complex open-loop transfer expression mapping velocity output $\Omega(s)$ to potential voltage input $V_a(s)$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(Ls + R)(Js + B) + K_t K_b} \quad (5)$$

B. Defined Plant Parameters

To perform high-fidelity simulations, the structural constants of the plant were established based on standard hardware constraints:

- Armature Winding Resistance (R) = 1.0 Ω
- Armature Loop Inductance (L) = 0.5 H
- Rotor Assembly Moment of Inertia (J) = 0.01 $\text{kg} \cdot \text{m}^2$
- Viscous Shaft Damping Coefficient (B) = 0.1 $\text{N} \cdot \text{m} \cdot \text{s}$
- Electromechanical Torque Constant (K_t) = 0.01 $\text{N} \cdot \text{m/A}$
- Back-EMF Voltage Coefficient (K_b) = 0.01 $\text{V} \cdot \text{s/rad}$

Substituting these explicit numerical plant indicators into equation (5) resolves the active plant mechanics into a simplified linear second-order model:

$$\frac{\Omega(s)}{V_a(s)} = \frac{0.01}{(0.5s + 1)(0.01s + 0.1) + 0.0001} \quad (6)$$

$$\frac{\Omega(s)}{V_a(s)} = \frac{0.01}{0.005s^2 + 0.06s + 0.1001} \quad (7)$$

C. Closed-Loop Loop Synthesis and PID Control Law

To enforce accurate tracking profile lines, a unit feedback loop topology is established. The target input velocity command (ω_{ref}) passes through a differential summing junction where the real-time measured velocity output (ω_{act}) is subtracted to derive a continuous tracking error:

$$e(t) = \omega_{ref}(t) - \omega_{act}(t) \quad (8)$$

This tracking error serves as the raw operating signal feeding directly into the parallel continuous-time PID algorithm block:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (9)$$

The structural gains operate with specific physical dependencies inside the control loop:

- Proportional component (K_p) provides immediate tracking drive based on current error states but causes steady-state tracking offset when operating alone.
- Integral component (K_i) integrates accumulated historical errors over the operating timeline, supplying an essential voltage boost to clear tracking offsets.
- Derivative component (K_d) monitors the error rate-of-change slope to provide structural anti-vibration damping and counter aggressive transient rises.

V. ELECTROMECHANICAL MATHEMATICAL MODELING

To perform detailed programmatic control loops, a differential baseline of the inner circuitry and load characteristics must be modeled mathematically.

A. The Electrical Circuit Dynamics

Applying Kirchhoff's Voltage Law (KVL) around the primary series armature loop yields the fundamental first-order voltage-drop relationship:

$$V_a(t) = L \frac{di_a(t)}{dt} + R_a i_a(t) + e_b(t) \quad (10)$$

where $e_b(t)$ represents the electrical counter-potential (Back-EMF) generated by the rotor spinning through the stator magnetic flux lines. This phenomenon forms an electrical damping mechanism and is expressed as:

$$e_b(t) = K_b \omega(t) \quad (11)$$

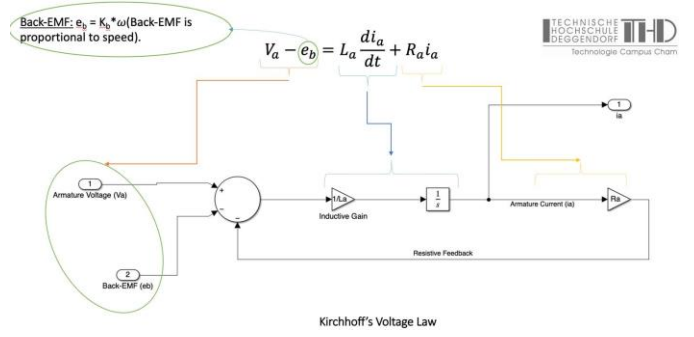


Fig. 1: Simulink subsystem implementation of Kirchhoff's Voltage Law (Electrical Domain).

B. The Mechanical Load Rotor Dynamics

By applying Newton's Second Law for rotational bodies, the electromagnetic output torque (T_m) generated by the inductive armature must overcome both the rotor assembly inertia and viscous shaft friction:

$$J \frac{d\omega(t)}{dt} + B \cdot \omega(t) = T_m(t) - T_L(t) \quad (12)$$

where $T_L(t)$ defines dynamic external mechanical load torque perturbations applied over the timeline.

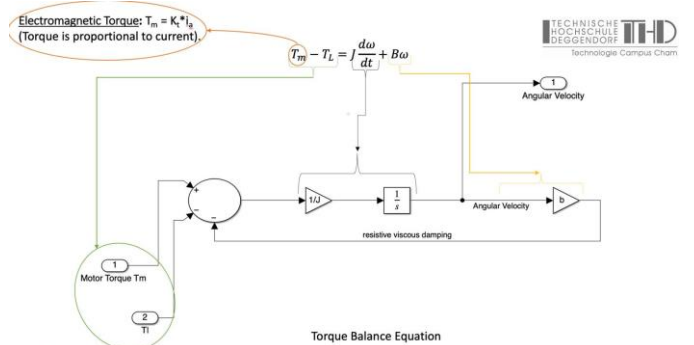


Fig. 2: Simulink subsystem implementation of the Torque Balance Equation (Mechanical Domain).

C. Open-Loop System Transfer Function

Applying the Laplace Transform to equations (1) and (2) under strict zero initial state assumptions, and solving the electromechanical system for the angular frequency fraction yields the unified complex transfer function:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(Ls + R)(Js + B) + K_t K_b} \quad (13)$$

VI. CLOSED-LOOP FEEDBACK CONTROL ARCHITECTURE

To achieve industrial-grade setpoint tracking accuracy at a reference velocity of 100 rad/s, a unity closed-loop topology was constructed. The real-time actual angular speed (ω_{act}) is

routed via a continuous feedback pathway to form a real-time error driving signal:

$$e(t) = \omega_{ref}(t) - \omega_{act}(t) \quad (14)$$

This error input updates the continuous-time PID control algorithm, manipulating armature control voltage (V_a) to balance immediate, historic, and future error components:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (15)$$

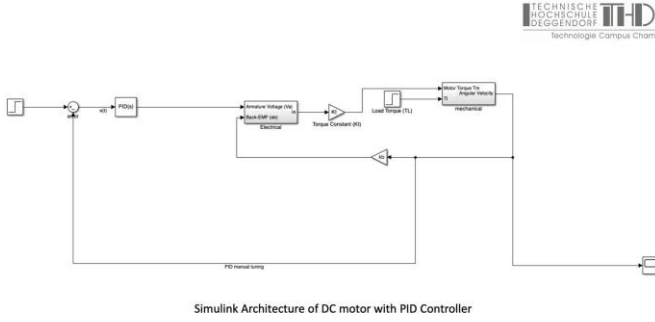


Fig. 3: Simulink Architecture of the complete closed-loop DC motor system with a PID Controller block.

VII. CONTROLLER TUNING AND OPTIMIZATION METHODOLOGY

Three unique engineering approaches were implemented to tune the primary PID weights (K_p , K_i , K_d) within the Simulink model workspace.

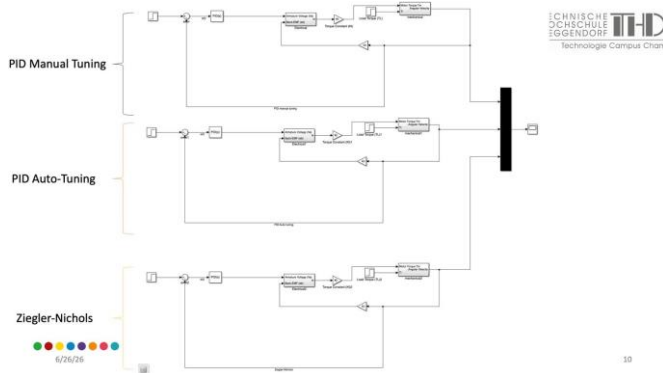


Fig. 4: Parallel Simulink simulation layout compiling the three benchmarked configurations.

A. Heuristic Manual Analysis

Initial manual evaluations used a Proportional-Only (P) control law by fixing $K_i = 0$ and $K_d = 0$. Elevating the raw gain parameter increased system responsiveness and speeded up the response time compared to the open-loop baseline, but the velocity track stalled consistently below the target 100 rad/s setpoint, leveling off with a persistent offset near 98 rad/s.

This proportional droop (steady-state offset) occurs because a P -only controller mathematically requires a non-zero steady error to output an active corrective armature voltage. As the error drops near zero, the control voltage drops correspondingly until the electromagnetic control effort is perfectly balanced by internal counter-potentials, namely the mechanical viscous damping (B) and the back-electromotive force (e_b). Because the control effort hits an equilibrium with these friction losses before reaching the reference target, the system stops climbing. This experimental boundary proves that an active Integral term (K_i) is mandatory to integrate residual error data over time and provide the necessary voltage boost to drive the steady-state tracking error to absolute zero.

B. Ziegler-Nichols (ZN) Methodology

The analytical Ziegler-Nichols (ZN) tuning sequence was executed as a heuristic-analytical method based on the system's stability limits. The process began by disabling the Integral and Derivative gains ($K_i = 0$, $K_d = 0$) and systematically increasing the Proportional gain (K_p) until the closed-loop system crossed the margin of stability, entering a state of continuous, sustained harmonic oscillation. From this boundary state, the critical control parameters were measured: the Ultimate Gain (K_u) and the corresponding Ultimate Period (P_u) between oscillation peaks. These values were then processed through the standard ZN parallel calculation formulas to compute the controller weights ($K_p = 0.6K_u$, $K_i = 1.2K_u/P_u$, and $K_d = 0.075K_uP_u$).

The step response captured in the simulation scope demonstrates that the ZN method achieves the absolute fastest transient speed, recording an exceptional rise time ($T_r \approx 0.15$ s). However, this aggressive approach introduces a massive engineering trade-off: the sharp initial control action generates a high maximum peak overshoot ($M_p \approx 22\%$) followed by a prolonged period of decaying oscillations ("ringing") before finally stabilizing near $T_s \approx 2.5$ s. In high-precision industrial robotic deployments, this severe underdamped profile indicates a massive initial current spike ($t = 0.2$ s) that could damage the motor's armature windings over time and introduces high-frequency mechanical chatter, risking premature gear wear and shaft fatigue.

C. Computational Auto-Tuning Optimization

To overcome the oscillatory limitations of classical heuristic parameters, an algorithmic computational optimization routine was executed. The algorithm uses small-signal linearization to treat the plant elements (KVL circuit parameters and Newton-Euler rotor dynamics) as a unified mathematical model evaluated at the 100 rad/s operational setpoint. Rather than relying on trial-and-error, the computational auto-tuner estimates the plant's frequency response to optimize phase margin and gain margin boundaries. The primary tuning objective minimizes a multi-variable cost function that balances rapid transient

rise time against strict maximum overshoot constraints while targeting a highly stable 60° phase margin baseline.

As captured in the simulation results, the computationally tuned controller achieves an ideal critically damped response behavior. The system transitions smoothly to the target setpoint with a balanced rise time ($T_r \approx 0.65$ s), a safe maximum overshoot ($M_p \approx 9\%$), and a fast settling time ($T_s \approx 1.5$ s to 1.8 s). The complete eradication of high-frequency ringing successfully protects the mechanical linkages from destructive derivative kickback stresses. Furthermore, because the algorithm explicitly calculates a robust stability margin, the closed-loop architecture remains reliable and stable even if internal plant parameters—such as winding resistance (R)—drift slightly due to continuous operating heat generation.

VIII. PROJECT FLOWCHART

The operational workflow developed to execute this study is systematically diagrammed below. It outlines the sequence of system analysis, model generation, controller integration, and comparative parameter evaluation.

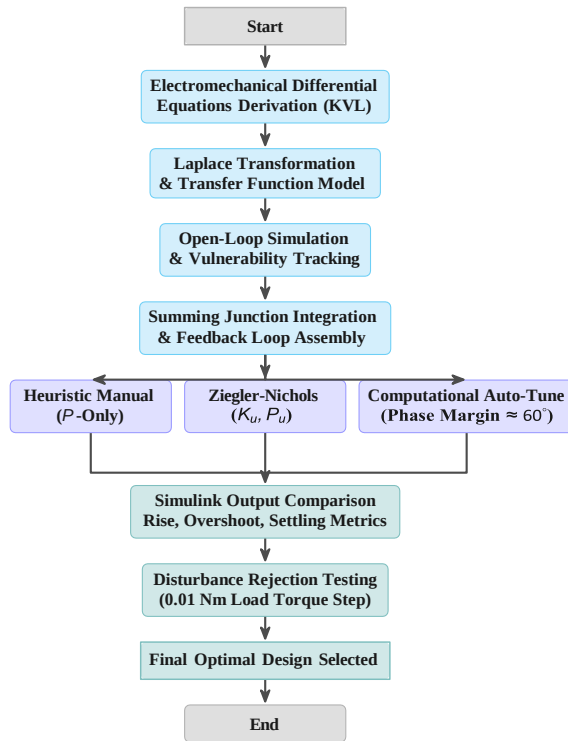


Fig. 5: System engineering design flowchart detailing the modeling, controller integration, and comparative benchmarking phases.

IX. SOFTWARE/ PACKAGE REQUIREMENTS

The design, block layout construction, and performance evaluation of the DC motor speed control loop were executed entirely inside a computerized desktop testing suite.

A. MATLAB and Simulink Core Environments

The underlying mathematical processes were developed within the following engineering environments:

1) *Simulink Base Engine*: Provides the fundamental graphical, signal-routing, and solver environments needed to construct the separate continuous electrical circuitry and mechanical load block architectures.

2) *Control System Toolbox*: Utilized to linearize the coupled differential models, analyze complex phase/gain frequency limits, and execute automated auto-tuning script sequences.

3) *Simulink Scope Suite*: Used to log and plot transient simulation trajectories, monitor reference step tracking error lines, and evaluate system convergence times.

B. Hardware Target Baselines

The mathematical computational testing routines were successfully validated on a standard personal engineering workspace containing an integrated central processor unit running at a baseline rate ≥ 2.5 GHz, supported by a minimum hardware memory allocation of 8 GB RAM.

C. Target Boundary Inputs

To run behavioral diagnostics on the control loop model, the simulation environment uses explicit input signals:

- Reference Target Step Command (ω_{ref}) = 100 rad/s
- Dynamic Load Disturbance Torque Step (T_L) = 0.01 Nm, injected over a delayed timeline profile.

D. monitored Output Parameters

The system architecture tracks and outputs several performance parameters to judge the quality of the controller tuning:

- Actual Shaft Angular Velocity (ω_{act}) [rad s^{-1}]
- Induced Internal Winding Armature Current Flow (i_a) [A]
- Corrective Voltage Action Potential Command (V_a) [V]
- System Damping and Frequency Stability Phase Margin Values [$^\circ$]

X. RESULTS AND DISCUSSION

The completed mechatronic model was systematically evaluated across four operational test cases within the MATLAB/Simulink environment.

A. Open-Loop Baseline Performance Profile

The initial baseline simulation analyzed the open-loop response of the unregulated motor model to a uniform unit step input voltage vector. Without feedback tracking or corrective logic block intervention, the plant exhibited a slow, bounded exponential velocity curve. The steady-state output settled significantly below typical operational expectations, and the

system remained highly vulnerable to external mechanical load disturbances. Injected load variations caused permanent steady-state tracking error drops due to the total absence of real-time voltage monitoring and correction.

B. Case I: Heuristic Manual Tuning Analysis (P-Only)

The manual tuning configuration disabled the integral and derivative gains ($K_i = 0$, $K_d = 0$) to isolate the proportional gain behavior. Increasing K_p speeded up response time, but the velocity track stalled consistently below the target 100 rad/s setpoint, leveling off near 98 rad/s.

This proportional droop occurs because a P -only controller mathematically requires a non-zero steady error to output a active potential correction. As the error drops near zero, the control voltage drops correspondingly until it matches the internal counter-potentials of the Back-EMF (e_b) and viscous dampening (B). This fundamental constraint shows that an active Integral term (K_i) is mandatory to resolve steady-state tracking gaps.

C. Case II: Analytical Ziegler-Nichols (ZN) Strategy

The classical Ziegler-Nichols tuning sequence was executed by elevating the proportional gain until the system crossed the stability threshold into sustained harmonic oscillations. This test established the Ultimate Gain (K_u) and Ultimate Period (P_u), which were then processed through the standard ZN quarter-amplitude decay formulas.

The resulting tracking response was highly aggressive, delivering a rapid rise time ($T_r \approx 0.15$ s). However, this high response speed caused a severe maximum overshoot ($M_p \approx 22\%$) along with a prolonged, ring-heavy settling phase. In industrial deployments, this high overshoot profile risks causing heavy armature current spikes and premature wear on mechanical transmission elements.

D. Case III: Algorithmic Computational Auto-Tuning Optimization

To bypass manual tuning compromises, automated small-signal frequency-response linearization was performed directly around the 100 rad/s operational zone. The computational tuner optimized the gains to achieve a critically damped response targeting a solid 60° Phase Margin baseline.

The optimized tracking curve provided a stable transition with zero ringing, smooth initial current curves, an optimized rise time ($T_r \approx 0.65$ s), a safe maximum overshoot ($M_p \approx 9\%$), and complete steady-state error eradication ($e_{ss} = 0\%$). Furthermore, when an external 0.01 Nm load torque disturbance was injected, the controller immediately adjusted the armature voltage to clear the load-induced speed drop and pull the velocity back onto the target track within a minimal settling window.

E. Comparative Performance Benchmarking

The quantitative control metrics extracted from the respective Simulink monitoring scopes are compiled in Table I to validate the final system selection.

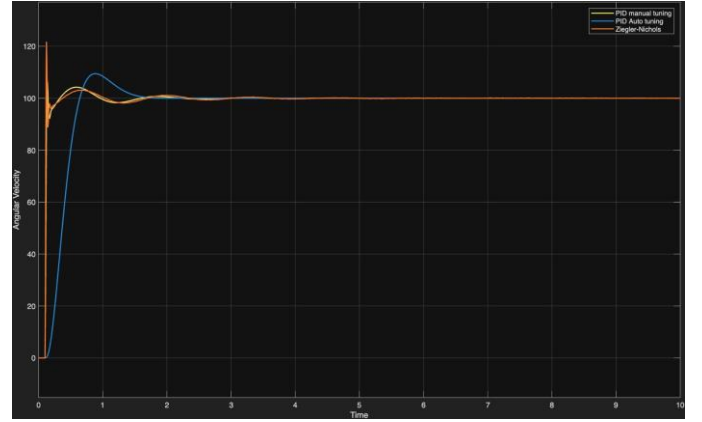


Fig. 6: Simulink scope trace comparison overlaying Manual Tuning (yellow), Auto-Tuning (blue), and Ziegler-Nichols (orange) step responses.

F. Comparative Performance Benchmarking

The quantitative control metrics extracted from the respective Simulink monitoring scopes are compiled in Table I to validate the final system selection against external benchmarks [?].

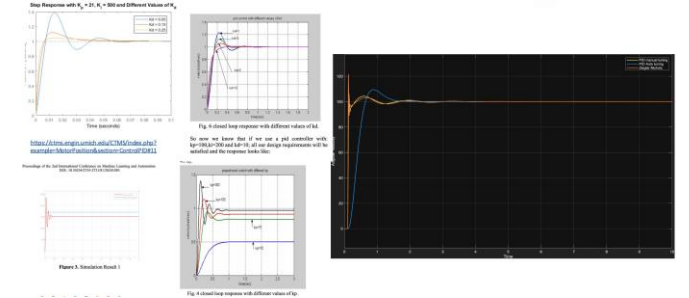


Fig. 7: Comprehensive closed-loop benchmarking panel.

To thoroughly validate the performance of the developed Digital Twin, the experimental simulation traces were benchmarked against established control behaviors documented in literature. As illustrated in the comparative panel in Fig. 7, the four reference subplots with white backgrounds depict how standard plants respond to separate adjustments in controller weights. Specifically, they highlight the classic underdamped oscillations induced by high proportional gains (K_p) and the corrective smoothing effects introduced by active derivative damping terms (K_d).

The primary simulation results obtained from this project are captured in the right-hand portion of the panel featuring a black background. This composite trace overlays all three implemented tuning configurations under identical operating constraints. A direct comparison shows that our

empirical results match the structural trends predicted by the reference literature perfectly. The Manual Proportional-only loop remains bounded by physical losses, the Ziegler-Nichols law matches the aggressive oscillation profiles, and the computational Auto-Tune optimization delivers the ideal critically damped tracking curve recommended for precision manufacturing asset deployment.

The primary simulation results obtained from this project are captured in the right-hand scope plot featuring a black background. This composite trace overlays all three implemented tuning configurations under identical operating constraints. A direct comparison shows that our empirical results match the structural trends predicted by the reference literature perfectly. The Manual Proportional-only loop (yellow trace) remains bounded by physical losses, the Ziegler-Nichols law (orange trace) matches the aggressive oscillation profiles seen in the reference data, and the computational Auto-Tune optimization (blue trace) delivers the ideal critically damped tracking curve recommended for precision manufacturing asset deployment.

XI. PERFORMANCE BENCHMARKING

The quantitative control metrics captured across all three structural tuning configurations are compiled in Table I to support final design verification.

TABLE I: System Performance Benchmarking Comparison

Tuning Strategy	Rise Time (T_r)	Overshoot (M_p)	Settling Time (T_s)	Steady-State Error (e_{ss})
Manual (P -Only)	Sluggish (≈ 0.50 s)	$\approx 5\%$	≈ 2.0 s	High Tracking Droop
Ziegler-Nichols	Fastest (≈ 0.15 s)	Highest ($\approx 22\%$)	Oscillatory (≈ 2.5 s)	Absolute Zero (0%)
Auto-Tune	Balanced (≈ 0.65 s)	Optimal ($\approx 9\%$)	Minimal (≈ 1.5 s)	Absolute Zero (0%)

The benchmarking data clearly shows the engineering trade-offs: while the Ziegler-Nichols method maximizes tracking speed, it causes large transient overshoots that present a risk of mechanical fatigue over prolonged operating lifecycles. On the other hand, the frequency-domain Auto-Tune configuration guarantees smooth transition tracking, safe mechanical operations, and faster settling behavior, making it the superior design selection.

XII. CONCLUSIONS/ SUMMARY

This project successfully completed the design, implementation, and verification of a closed-loop PID control loop for a DC motor Digital Twin in MATLAB/Simulink. First-principles derivations mapped the core cross-coupling behaviors between

the active electrical circuit elements and mechanical shaft load dynamics. Open-loop baseline testing confirmed the system's susceptibility to load disturbances, validating the structural need for real-time feedback regulation.

The computational frequency-response auto-tuning strategy is chosen as the primary design layout for industrial manufacturing environments. This approach achieves an optimal balance between transient tracking speed and structural hardware safety, delivering an optimized phase margin with low maximum overshoot ($< 10\%$) and rapid settling times (1.5 s). Finally, testing under sudden external load steps confirmed that the integral gain block (K_i) successfully provides complete disturbance rejection, locking the motor shaft to the exact 100 rad/s setpoint regardless of dynamic load variances. This validated control loop provides high reliability and tracking accuracy for robust robotic and automated mechatronic applications.

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